25900014 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION, MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCBIF100 - BIOMOLECULES AND MOLECULAR VISUALIZATION

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer Questions

Answer any **5** questions

Each question carries **2** marks

1. Comment on peptide bond? [U] / [CO1]
   It is a crucial chemical bond in biology, specifically in the formation of proteins and polypeptides.
2. Name any two types of RNAs found in living beings. [A] / [CO2]
   messenger RNA (mRNA) this type of RNA carries genetic information from DNA to the ribosome and transfer RNA (tRNA) responsible for transporting specific amino acids to the ribosome.
3. What are the three major forms of DNA secondary structure? [K] / [CO2]
   The three major forms are A-DNA, B-DNA, and Z-DNA. These forms are all double-helical structures, but differ in their helical parameters, groove dimensions, and base pair orientations.
4. Chart the main types of biological databases. [A] / [CO3]
   Biological databases can be broadly classified into Primary databases store original experimental data. Secondary databases curate and analyze data from primary sources. Composite databases combine data from various sources.
5. Operate a pairwise sequence alignment tool for comparing two sequences. [An] / [CO4]
   BLAST - input two sequences, and the tool will identify regions of similarity between them. The tool will then display the alignment, highlighting matching and non-matching regions.
6. Write one advantage of using Swiss PDB Viewer for molecular visualization. [A] / [CO5, CO6]
   It provides a user-friendly interface allowing to analyze several proteins at the same time.
7. Discover which organization developed Map Viewer. [A] / [CO5, CO6]
   It was developed by the National Remote Sensing Centre (NRSC) of the Indian Space Research Organization (ISRO).

[2x5 = 10]

Part B

Short Essay Questions

Answer any **3** questions

Each question carries **5** marks

8. How would you explain the role of glycogen in physical activity? [A] / [CO1]
Glycogen is a stored form of carbohydrate found mainly in the liver and muscles.(1 mark) During physical activity, the body breaks down glycogen into glucose, which is used by muscle cells to produce energy. This process is especially important during moderate to high-intensity exercise, where the demand for energy increases rapidly. Muscle glycogen provides a quick and accessible energy source, helping to maintain performance and delay fatigue. As exercise continues, glycogen stores become depleted, which can lead to a drop in energy levels and reduced endurance. Therefore, adequate glycogen storage is essential for optimal athletic performance and recovery.(4 marks)

9 What are the three main types of RNA, and what are their roles? [U] / [CO2]
RNA (ribonucleic acid) is a single-stranded molecule involved in the process of coding, decoding, regulating, and expressing genes.(1 mark) Types - mRNA (messenger RNA) – Carries genetic information from DNA in the nucleus to the ribosomes, where proteins are synthesized..tRNA (transfer RNA) – Brings the correct amino acids to the ribosome during protein synthesis by matching its anticodon to the mRNA codon and rRNA (ribosomal RNA) – A structural component of ribosomes that helps catalyze the formation of peptide bonds between amino acids during protein synthesis. (4 marks)

10. Explain the importance of MSA in identifying conserved protein domains. [An] / [CO4]
Multiple Sequence Alignment (MSA) is important for identifying conserved protein domains because it compares amino acid sequences from different proteins or species. By aligning these sequences, MSA highlights regions that are highly similar or unchanged (conserved), suggesting they have important structural or functional roles. Conserved domains often indicate shared ancestry and are crucial for the protein's function, such as binding or catalytic activity. Identifying these domains helps in predicting protein function, understanding evolution, and guiding drug design or genetic research. (5 marks)

11. Interpret ribbon model and its applications. [A] / [CO5, CO6]
It is a graphical representation of a protein's 3D structure. It uses ribbons to show the backbone of the protein, highlighting key features. This simplified model helps visualize the overall folding and arrangement of the protein without showing every atom.(2.5 marks) Applications Understanding how proteins fold and function,Identifying active sites or binding regions,Comparing structural similarities between proteins ,Supporting drug design by revealing how molecules interact with proteins.(2.5marks)

12. Explain molecular graphics and its types. [A] / [CO5, CO6]
It is the use of computer-generated visualizations to represent the structure, shape, and interactions of molecules, especially large biological macromolecules like proteins, DNA, and ligands. It helps researchers understand molecular function, predict interactions, and support drug design.(2.5 marks) Types - Wireframe model, Ball-and-stick model, Space-filling model, Ribbon model.(2.5 marks)

[5x3 = 15]

Part C

Long essay Questions

Answer any **1** questions

Each question carries **10** marks

13. What is the difference between monounsaturated and polyunsaturated fatty acids? Give examples. [U] / [CO1]

Monounsaturated fatty acids and polyunsaturated fatty acids are two types of unsaturated fats, and while they share some similarities, they differ in their structure, sources, and health benefits. Monounsaturated fatty acids contain one double bond in their carbon chain. This makes MUFAs typically liquid at room temperature. The chemical structure of a MUFA is characterized by a long hydrocarbon chain with one point of unsaturation. Example - Oleic acid, Palmitoleic aci, Erucic acid. Polyunsaturated fatty acids contain more than one double bond in their carbon chains. These multiple double bonds prevent the molecules from packing tightly together, keeping them liquid at room temperature and even at lower temperatures.Example - Alpha-linolenic acid, Eicosapentaenoic acid, Docosahexaenoic acid (5 Marks) Differences - Chemical Structure:MUFAs: One double bond in the carbon chain and PUFAs: More than one double bond in the carbon chain.Sources: MUFAs: Olive oil, avocados, nuts, and seeds and PUFAs: Fish (omega-3), flaxseeds, walnuts, soybean oil (omega-6), and sunflower oil.Health Effects:MUFAs: Lower LDL cholesterol, improve insulin sensitivity, and have anti-inflammatory effects and PUFAs: Omega-3 fatty acids are anti-inflammatory and protect heart health, while omega-6 fatty acids can be beneficial but may promote inflammation if consumed in excess relative to omega-3.Physical State: MUFAs: Typically liquid at room temperature and PUFAs: Also liquid at room temperature and even colder temperatures.(5 marks)

14. Compare sequence databases and structure databases with examples. [A] / [CO3]

The sequence databases and structure databases are both essential tools for storing and organizing biological information.Sequence databases store information about the linear sequence of nucleotides or amino acids, structure databases contain data about the three-dimensional shapes of biomolecules, especially proteins and nucleic acids.(3 marks) Sequence databases store nucleotide or protein sequences, which are linear, and are primarily used for sequence alignment, gene prediction, and evolutionary studies. They focus on the order of nucleotides or amino acids in a sequence. Examples of sequence databases include GenBank, UniProt, and EMBL. These databases rely on sequence-based techniques, such as DNA sequencing, to gather data and are commonly applied in genomic studies, functional annotations, and mutation analysis. structure databases contain 3D coordinates of atoms in molecules, providing detailed information about molecular shapes, folding, and interactions. They are used for visualizing and analyzing the three-dimensional structures of biomolecules. Examples of structure databases include the Protein Data Bank (PDB), ModBase, and SCOP. Structure databases rely on experimental methods like X-ray crystallography and NMR spectroscopy, as well as computational modeling, to determine 3D structures. These databases are mainly applied in drug design, protein interaction studies, and molecular modeling.(7 marks)

15. Explain MolScript, and what is its primary function in molecular visualization. [S] / [CO5, CO6]

MolScript is a software tool used for generating high-quality molecular graphics and visualizations, particularly for proteins, nucleic acids, and other biomolecules. It is widely used in structural biology for creating publication-quality figures that illustrate the three-dimensional structures of molecules.(2 marks) MolScript's primary function is to create vector-based graphics that represent the 3D structure of biomolecules. It allows users to produce images of molecular structures in various styles, such as ribbons, sticks, and spheres, providing a clear and effective way to visualize complex molecular details. The tool can depict molecular features such as secondary structures (e.g., alpha-helices, beta-sheets), interactions between molecules, and the

overall arrangement of atoms, helping researchers to understand and present molecular structures more easily. It is particularly useful for illustrating protein folds, the positioning of functional groups, and the interactions between proteins and ligands, making it an essential tool for visualizing and communicating structural data in molecular biology.(8 marks)

[10x1 = 10]